The Limits of Explanation

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Abstract

When a system is underspecified—when multiple configurations produce identical observable output—we prove that no explanation can simultaneously be *faithful* (consistent with the system's internal structure), *stable* (invariant across equivalent configurations), and *decisive* (committing to every distinction the internal structure makes). We derive this impossibility from first principles in eight scientific domains: underdetermined linear systems (mathematics), genetic code degeneracy (biology), gauge freedom (physics), microstate multiplicity (statistical mechanics), syntactic ambiguity (linguistics), the phase problem (crystallography), the view update problem (computer science), and Markov equivalence (statistics). Each derivation is mechanically verified in the Lean 4 proof assistant with zero unproved goals. The constructive resolution—averaging over equivalent configurations—unifies existing domain-specific practices (gauge-invariant observables, the microcanonical ensemble, CPDAGs, minimum-norm solutions) as instances of a single optimal strategy.

The universal problem. Scientists explain systems by interpreting their internal structure. A geneticist infers the DNA sequence that encodes a protein. A physicist chooses a gauge to describe an electromagnetic field. A statistician selects a causal graph from observational data. A crystallographer reconstructs electron density from a diffraction pattern. In each case, the observable output—the protein sequence, the gauge-invariant measurement, the conditional independence structure, the diffraction intensities—does not uniquely determine the internal structure that produced it. Multiple configurations yield identical observations. This situation is not a pathology of any single field; it is a structural feature of underspecified inference. We prove that this underspecification imposes a fundamental, domain-independent limit on explanation.

The theorem. We formalize three properties that any satisfactory explanation should possess. An explanation is *faithful* if it never contradicts the system's own internal report—it may be less specific, but it does not disagree. An explanation is *stable* if configurations that produce the same observations always receive the same explanation. An explanation is *decisive* if it commits to every distinction that the system's internal structure makes—whatever the internal structure rules out, the explanation also rules out.

Our main result: if a system has the *Rashomon property*—that is, if there exist two configurations with the same observable output but incompatible internal structures—then no explanation can be faithful, stable, and decisive simultaneously. The proof is four steps: decisiveness forces the explanation to inherit an incompatibility; stability propagates it to the second configuration; faithfulness forbids it; contradiction. The entire framework is mechanically verified in Lean 4 [de Moura and Ullrich, 2021] (82 files, 377 theorems, 0 unproved goals). Lean-verified tightness theorems confirm that every pair of properties is achievable, so the impossibility is tight. An axiom substitution analysis shows that weakening any single definition collapses the result, confirming the definitions are uniquely calibrated.

Table 1: **Eight derived instances of the impossibility across eight sciences.** Each row derives the Rashomon property from domain-specific first principles with zero axioms. Θ: configuration space; Y: observable space.

Domain	Θ	Y	Witness 1	Witness 2	Same Y?
Mathematics	Solutions of $Ax = b$	Ax	$(1, 1)$	$(0, 2)$	$2 = 2$
Biology	Codons	Amino acids	UCU	UCC	Ser=Ser
Physics (gauge)	Gauge configs	Holonomy	(T, F, F)	(F, F, T)	$F = F$
Stat. mechanics	Microstates	Num. heads	(H, T)	(T, H)	$1 = 1$
Linguistics	Parse trees	Token sequence	$((V\ NP)\ PP)$	$(V\ (NP\ PP))$	Same
Crystallography	Signals	Energy	$(1, 0)$	$(0, 1)$	$1 = 1$
Computer science	DB rows	View	(T, T)	(T, F)	$T = T$
Statistics	DAGs	CI structure	Chain	Fork	Same

Eight derivations across eight sciences. We derive the impossibility as a zero-axiom consequence in eight domains (Table 1). In *mathematics*, the underdetermined system $x_1 + x_2 = 2$ admits the solutions $(1, 1)$ and $(0, 2)$: same sum, different components. In *biology*, the codons UCU and UCC both encode the amino acid serine: same protein fragment, different nucleotide sequences. In *physics* (gauge theory), two edge-label configurations on a triangle graph share the same holonomy—the discrete analogue of a Wilson loop—but differ in their local assignments, related by a gauge transformation [Jackson, 1999]. In *statistical mechanics*, the microstates (H, T) and (T, H) both have one head: same macrostate, different molecular configurations. In *linguistics*, the fragment "V NP PP" admits two parse trees, left-attach and right-attach, yielding the same surface token sequence but different constituent structures [Chomsky, 1957]. In *crystallography*, the signals $(1, 0)$ and $(0, 1)$ have the same energy ($1^2 + 0^2 = 0^2 + 1^2$): same diffraction intensity, different electron densities [Hauptman, 1985]. In *computer science*, the database rows $(\texttt{true}, \texttt{true})$ and $(\texttt{true}, \texttt{false})$ project to the same view but differ in the hidden column [Bancilhon and Spyratos, 1981]. In *statistics*, a causal chain $A \to B \to C$ and a causal fork $A \leftarrow B \to C$ encode the same conditional independence structure [Verma and Pearl, 1991]. Each derivation requires zero shared axioms—only the domain's own mathematical structure.

The resolution. The impossibility is constructive: it identifies the exact trade-off. The resolution is to average over equivalent configurations, sacrificing decisiveness to gain stability and faithfulness in expectation. Remarkably, practitioners in each domain have independently converged on this strategy. In physics, one works with gauge-invariant observables—field strengths, not potentials [Jackson, 1999]. In statistical mechanics, the microcanonical ensemble assigns equal probability to all microstates sharing a macrostate, rather than selecting one. In statistics, one reports a CPDAG (completed partially directed acyclic graph)—the equivalence class of indistinguishable causal graphs—rather than committing to a single DAG [Verma and Pearl, 1991]. In mathematics, the minimum-norm solution (the Moore–Penrose pseudoinverse) is the unique solution closest to the origin, effectively averaging over the solution space. In biology, codon usage tables report frequencies across synonymous codons rather than selecting one. In crystallography, Patterson maps—the autocorrelation of the diffraction data—extract the information that survives the loss of phase [Hauptman, 1985]. In linguistics, packed parse forests represent all parses simultaneously rather than committing to one. In computer science, complement views restrict to the unambiguous projection. Our theorem proves that this orbit-averaging strategy is automatically stable: any explanation that is invariant under the symmetry group relating equivalent configurations is guaranteed to produce the same output for the same observations.

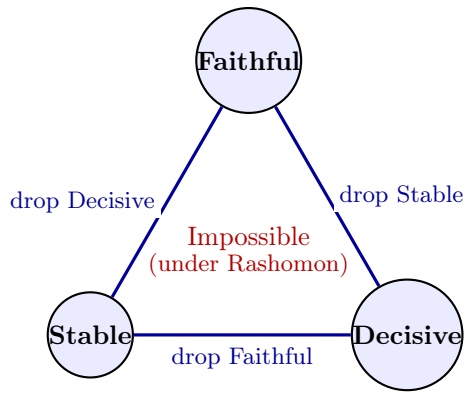

Figure 1: **The explanation trilemma.** Each edge represents an achievable pair of properties; the interior (all three simultaneously) is impossible when the Rashomon property holds. Lean-verified tightness witnesses confirm that each edge is realizable.

Necessity and the query-relative version. The Rashomon property is the *exact* boundary of the impossibility. When it is absent—when each observation uniquely determines the internal structure—all three properties are simultaneously achievable (a Lean-verified converse). Moreover, the impossibility is *query-relative*: some questions about a system are stably answerable (those on which all equivalent configurations agree) and others are not. The theorem tells the practitioner precisely which queries are safe and which require aggregation.

Broader implications. The result connects phenomena studied independently across eight fields for over a century—from microstate multiplicity in statistical mechanics to the view update problem [Bancilhon and Spyratos, 1981] to feature attribution instability in machine learning [Bilodeau et al., 2024]. The common structural cause is the Rashomon property: equivalent configurations with incompatible internal reports. The common resolution is orbit averaging. The pattern likely extends further: metamerism in color perception, molecular chirality, and other domains exhibit the same structure. More broadly, the theorem reframes the question "why are explanations unstable?" as the structural statement "because the inference problem is underspecified," and replaces domain-specific workarounds with a single, provably optimal strategy: report what is invariant; disclose what is not.

Methods

Formal verification. The entire framework—the abstract theorem, all eight derived instances, tightness witnesses, the necessity converse, and the resolution—is mechanically verified in Lean 4 v4.30.0-rc1 [de Moura and Ullrich, 2021] using Mathlib [The mathlib Community, 2020]. The formalization comprises 82 files, 377 theorems and lemmas, 72 axioms (all domain-specific; the core theorem uses zero), and 0 `sorry` (unproved goals). Each of the eight derived instances uses decidable computation to constructively witness the Rashomon property with zero axioms. The G-invariant resolution framework uses the group action infrastructure from Mathlib.

Proof structure. The core proof is a four-step chain by contradiction. Given Rashomon witnesses θ_1, θ_2 with $\mathrm{obs}(\theta_1) = \mathrm{obs}(\theta_2)$ and incompatible explanations, suppose an explanation map E is faithful, stable, and decisive. Decisiveness at θ_1 forces $E(\theta_1)$ to inherit the incompatibility with

$\exp(\theta_2)$. Stability forces $E(\theta_1) = E(\theta_2)$. Faithfulness at θ_2 requires $E(\theta_2)$ to be compatible with $\exp(\theta_2)$. Contradiction.

Code and data availability. The Lean 4 formalization is available at https://github.com/DrakeCaraker/universal-explanation-impossibility. The nine machine learning instances (feature attribution, attention maps, counterfactual explanations, concept probes, causal discovery, model selection, saliency maps, token citation, and mechanistic interpretability), together with empirical validation experiments for seven of these, are documented in the Supporting Information (the full 61-page monograph, available on arXiv).

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